

v4 continuous carrier-frequency-offset report

At the frozen synthetic offsets of $\pm 0.125\%$ of the nominal carrier, the existing nominal-template decoder passed **0 of 12,000** noisy frames in each direction at sample-noise SD 0.45 ; the two-sided 95% Wilson upper bound was about **0.032%**. At $\pm 0.0625\%$, pass share was **53.17%** (95% Wilson interval 52.27–54.06%) for negative CFO and **52.32%** (51.42–53.21%) for positive CFO. With zero CFO it passed **11,365/12,000 (94.71%; 94.29–95.09%)**, agreeing with the v3 calibration scale.

This is a continuous phase-ramp experiment, distinct from v2' s fixed per-segment phase rotation. Communications references describe carrier-frequency offset as linearly accumulated phase over time and treat it separately from sampling-clock offset; carrier recovery texts likewise model receiver local-oscillator frequency mismatch (🔥 [MathWorks CCSDS optical-link example](#), 🧑 [Stanford Fundamentals of Synchronization](#)). Those sources motivate the impairment form, **not** the synthetic offset magnitudes chosen here.

The frozen protocol used six existing 32-value vectors, a two-sided grid from -0.05 to $+0.05$ extra cycles per 64-sample symbol (0 to $\pm 1.25\%$ of the four-cycle nominal carrier), exact symbol timing, clean controls, and IID Gaussian sample noise at SD 0.45 . At 2,000 frames per vector and offset, the stochastic run covered **132,000 input frames** with seed 20261006 . CFO phase accumulates continuously over the frame; it does not reset at symbol boundaries. The complete protocol and pre-run runner hash are recorded in the [protocol lock](#). The run' s JSON, CSV, and plot were saved; after inspecting the initial plot, only legend placement was adjusted and the figure was regenerated from the **unchanged saved result data**, without resampling.

The **existing decoder** was run unchanged and was not given CFO correction. A separate **oracle** knew the exact frequency offset, frame start, and symbol timing, and projected onto the matching template; it is an analysis-only upper-bound reference, not a replacement decoder. Across the full grid its noisy pass shares ranged from **94.37% to 94.99%**, near its closed-form predictions. Every nominal-decoder prediction fell inside its observed pooled Wilson interval. One of eleven oracle predictions lay just outside its interval; the largest oracle empirical/theory gap was **0.41 percentage points** at $+1.25\%$ CFO. At zero CFO the nominal and oracle use the same template and produced identical observations. The old always-zero method passed **0/6** fixed vectors and remains only a weak sanity floor, not a meaningful competing decoder.

For clean frames, the existing decoder passed **6/6 vectors** at zero and at $\pm 0.0625\%$ CFO, but **0/6** at $\pm 0.125\%$ and all larger tested magnitudes. The exact-offset oracle passed **6/6** at every clean CFO value. In the noisy transition region, per-vector nominal pass shares varied from **17.45% to 73.55%** at -0.0625% , and from **14.80% to 73.65%** at $+0.0625\%$; the pooled

intervals therefore describe Monte Carlo variation conditional on the fixed vector panel, not variation over a population of possible inputs. The analytic frame probabilities use each receiver's derived projection bias and Gaussian noise variance. One zero-pass result does not establish a physical failure threshold.

This remains a **dimensionless synthetic software test**, not a measured channel model or evidence of physical validity. Its abrupt outcome depends on the selected waveform, frame length, residual tolerance, fixed vectors, and synthetic CFO grid. The 39-test project suite passed after the run, the v4 JSON manifest validates, and all pre-run hashes for protected v1–v3 sources, tests, reports, and results match.

- [v4 runner](#)
- [v4 tests](#)
- [Protocol lock](#)
- [JSON results and protocol](#)
- [CSV results](#)
- [Summary plot](#)
- [Pre-run protected-file checksums](#)